Does Deliberation Debias? Exogenous Reasoning Effort and Behavioural Finance Bias in Large Language Model Portfolio Advisors

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Abstract

Language models are increasingly asked for advice about financial positions people already hold, and they carry measurable behavioural bias. Whether deliberation removes it is unsettled, because the available evidence either compares different models, so that capability and deliberation move together, or infers deliberation from the model's own account of its reasoning, produced by the same forward pass as the answer. We instead set the reasoning budget exogenously through the API, at three levels, before the model sees the item, and measure nine families of normatively irrelevant cue against matched pairs that differ only in that cue. Across 9,720 responses from three models the disposition effect runs opposite to the human direction: pooled, these models are more willing to sell at a loss than at a gain, $+0.239$ $[+0.148, +0.331]$ on the categorical measure and $+13.4$ $[+8.8, +18.0]$ percentage points of the position, where the pre-registered prediction was negative, though the reversal is carried by the two smaller models and appears in the strongest only as effort rises. Reasoning effort reduces the absolute effect in all nine qualifying cases, against a shuffled-label null of 0.0125 rather than the 2^{-9} a count would suggest, and it does so by compressing the spread of bias magnitudes across families: the mean falls detectably in the strongest model alone, and the compression itself is robust in the weakest, where no shuffle in 2,000 reproduces it, and inseparable from sampling error in the strongest. The exogenous manipulation is what makes this visible. Measured the observational way on the same responses, with output tokens standing in for displayed deliberation, more reasoning goes with *more* bias between items and with nothing at all within an item. We also report that `temperature` is unavailable on these models, which makes the fixed-temperature protocol used throughout this literature unexecutable, and give a repeated-sampling replacement.

1 Introduction

Language models now sit inside the tools through which ordinary people make financial decisions, and are increasingly asked not for a price or a definition but for a recommendation about a position someone already holds. That they carry measurable behavioural bias is the premise here rather than the finding: human dual-process failures reproduce in them (Hagendorff et al., 2023), thirty biases appear across twenty models (Malberg et al., 2025), and preference-based responses grow less rational as models advance (Bini et al., 2026).

Whether deliberation removes any of it is unsettled. Reasoning models are less prone to anchoring (Takenami et al., 2025) but no better protected in clinical vignettes (Kim et al., 2025); the effort parameter has been reported to set a ceiling on generation rather than to drive how reasoning is allocated (Hu and Wang, 2026); and an indicator for analytical reasoning raises rational responses by 40% (Bini et al., 2026). Each either compares different models, so that capability and deliberation move together, or measures something other than a decision, or relies on the model's own account of its reasoning. The reasoning budget has not been manipulated as a treatment against a behavioural finance bias. That is this paper.

Across 9,720 responses the disposition effect runs opposite to the human direction, reasoning effort reduces the absolute effect in all nine cases that qualified for the pre-registered test, and within a fixed effort level the association between generated length and bias is positive

rather than negative.

The 40% figure of Bini et al. (2026) is the sharpest case. Their indicator is the model's own characterisation of its reasoning, emitted in the same response as the answer it describes, so a model that answers well and then calls its own reasoning analytical produces the association with no causal path from deliberation to accuracy. A budget set before the model sees the item is independent of what it goes on to say.

That distinction can be measured here rather than argued, because the data holds both variables: the budget the experimenter set and the tokens the model chose to spend. Section 4.5 substitutes the second for the first on the same responses and the answer changes sign.

Our contributions are:

- The first measurement, to our knowledge, of behavioural finance bias under an exogenous, graded manipulation of reasoning effort.
- A matched pair battery of nine biases in advisory framing with realistic portfolio state, two of them absent from prior work, released in full.
- A distributional protocol, forced by the removal of sampling parameters from the API: repeated sampling and bootstrap intervals in place of single deterministic draws, with within-item entropy reported as a result.
- A test of what the exogeneity buys. Substituting an endogenous measure of displayed reasoning for the manipulated budget, on the same responses, reverses the sign.
- A pre-registered plan committed before the first API call, with the prompts, raw responses and analysis code.

2 Related Work

2.1 Behavioural biases in human investors

The biases tested here are documented regularities in human investors and the design borrows their operationalisations rather than inventing new ones: prospect theory (Kahneman and Tversky, 1979) and framing (Tversky and Kahneman, 1981), the disposition effect (Shefrin and Statman, 1985) as measured in brokerage data by Odean (1998), mental accounting (Thaler, 1985), myopic loss aversion (Benartzi and Thaler, 1995) and home bias (French and Poterba, 1991). Barber and Odean (2000) supply the reason to measure them in an advisor at all: they cost realised returns. Appendix J maps each finding to the family built from it and supplies the human baselines of Appendix O, a comparison that is indicative rather than like for like.

2.2 Behavioural bias in language models

That language models exhibit behavioural biases is established, and this paper does not re-establish it. Models have served as experimental subjects for cognitive tasks (Binz and Schulz, 2023; Hagendorff et al., 2023), as simulated economic agents (Horton et al., 2023) and as stand-ins for survey populations (Argyle et al., 2023; Aher et al., 2023). Three audits set the state of the art: thirty biases across twenty models (Malberg et al., 2025), sixteen cognitive-psychology questions across twelve models (Bini et al., 2026), and eight behavioural parameters over 24,000 agent-scenario pairs (Yee and Koh, 2026). Closest to this setting, Lee et al. (2025) measure sector, size and momentum preferences and show them escalating into confirmation bias.

What none of these do is ask a model whether to act on a position it already holds. The sixteen questions of Bini et al. (2026) are abstract gambles and general-knowledge items, with no disposition effect, mental accounting, advisor attribution, price path or brand salience. Malberg et al. (2025) do test the disposition effect, as a relative preference between two owned assets on a 7-point scale. In neither is a cost basis attached to a single position, and in neither is the model asked for a recommendation. That decision, whether to sell a position whose purchase price is stated, is the one the human field studies score, and it is absent from the model-side literature.

2.3 Deliberation and debiasing

Whether thinking harder makes a model less biased is an open disagreement, and the papers on each side are not testing the same thing.

Takenami et al. (2025) find reasoning models less prone to anchoring than non-reasoning

models in price negotiations, and attribute it to longer chains of thought. Kim et al. (2025) find that reasoning does not reliably protect against bias in clinical vignettes and that in one model it increased vulnerability. The first compares different models rather than varying deliberation inside one, so model capability and reasoning length are confounded.

Bini et al. (2026) come closest to an answer and are the paper this design is built against. Their mediation analysis reports that an indicator for analytical reasoning, which they label system 2 thinking, increases rational responses by 40% on preference-based questions. Their prompt asks the model for its choice, a confidence level, a short explanation and its reasoning type in one response, which is where the identification problem of Section 1 enters. Their one exogenous lever, role priming, moves that same indicator by 11.4 percentage points, so the large number belongs to the correlational mediator rather than to anything the experimenter set. Their treatments are prompt level throughout, and their own summary is that debiasing remains a challenging task; Appendix K gives those results in full.

The budget removes that problem: the treatment is assigned rather than reported, graded rather than binary, and invisible to the model's own account of itself.

Two papers manipulate the effort setting itself, and neither measures a financial decision. Hu and Wang (2026) run three levels on two GPT-OSS models and conclude that the parameter sets an upper budget on generation rather than driving how reasoning is allocated, with Bayes factors favouring the null. Their outcome is the alignment between generation length and human reaction times, not whether a decision changes, so it is pre-registered here as the competing hypothesis H10-null rather than treated as contradicted. Piya (2026) report instead that accuracy falls at medium effort, a reason to report the full profile and not only `low` against `high`.

No paper found manipulates the reasoning budget exogenously inside a fixed model and measures the change in a behavioural finance bias, which is what this paper does. Lee et al. (2025), the closest work in this domain, report themselves that their design is weak for reasoning models, and their action set of buy or sell with no hold makes the response ceiling of Appendix L unobservable by construction.

3 Method

3.1 Matched pair construction

Each item is a pair of prompts $(p_{\text{base}}, p_{\text{treat}})$ stating the same current price, share count, position weight, portfolio size and fundamentals block, and differing only in one manipulated cue. Under standard finance theory the normatively correct recommendation is identical across the pair, so

$$\Delta = \mathbb{E}[s \mid p_{\text{treat}}] - \mathbb{E}[s \mid p_{\text{base}}]$$

is the measured bias, where $s \in \{0, 1, 2, 3\}$ is the sell tendency. The same difference is computed on a continuous co-primary measure, the percentage of the position the model recommends selling, reported alongside s throughout.

One pair is reproduced in full at the start of Appendix V, which prints one pair from every family and the manipulated cue in all 108.

The first pilot was nearly fatal and the fix is part of the method. On stimuli whose fundamentals make neither buying nor selling obviously correct, the model returned `HOLD` in 93% of 108 responses, with no `BUY_MORE` and no `SELL_ALL`, so the dependent variable had no variance for any cue to move. Two remedies failed: dropping the normative instruction left `HOLD` at 83%, and a continuous target weight returned the current weight exactly in 28 of 36 responses. The cause was the stimulus. Fundamentals balanced so that neither buying nor selling is right also make leaving the position alone right, and then the cost basis genuinely is irrelevant. Every item therefore carries decision pressure, a cash need of 10% of the account in both arms, in a tax-exempt account so that realising a loss carries no tax advantage. `HOLD` fell to 71%. Appendix L gives the pilot numbers.

3.2 Bias families

Nine families are listed in Table 1. Each is a matched pair differing in one cue and identical in everything bearing on the decision, so the normatively correct recommendation is the same in both arms and a difference between them is a violation of invariance rather than

Family	Cue varied	Pred.	Status
Disposition	gain vs loss	−	conf.
Framing	gain vs loss frame	2-sided	conf.
Anchoring	low vs high anchor	−	conf.
Home bias	domestic vs foreign	+	conf.
Action bias	licence to do nothing	−	conf.
Mental acct.	long vs discretionary	+	conf.
Attribution	inherited vs self	−	conf.
Price path	fell-rose vs rose-fell	+	expl.
Brand salience	anonymous vs ticker	2-sided	expl.

Table 1: The nine bias families. Predicted direction is the sign of the mean sell tendency under the treatment arm minus the baseline arm, fixed in `analysis/directions.py` before the run. Advisor attribution and price path are original to this study. Table 4 adds the normative reason the recommendation should not change in each.

a response to information. Seven are confirmatory. Price path is exploratory because the route a price took is weak evidence rather than none, and brand salience because a real ticker brings the model's own prior knowledge; the two are reported separately and never pooled with the confirmatory families.

3.3 Companies and randomisation

Company profiles are synthetic, because a real ticker would carry the model's own training knowledge into every manipulation; real names appear only in the brand salience family, which is pre-registered as exploratory for that reason. Twelve profiles span sectors and are mixed on fundamentals, so neither buying nor selling is obviously correct, which is what makes the decision pressure of Appendix L necessary. Appendix I gives the randomisation.

3.4 Models, effort, and sampling

The three models are `claude-haiku-5-5`, `claude-sonnet-5-5` and `claude-opus-5-5`, each at `low`, `medium` and `high` reasoning effort, 108 pairs per cell and five trials per prompt, giving $3 \times 3 \times 108 \times 2 \times 5 = 9{,}720$ responses. All were collected through the batch API on 2026-10-26. Model identifiers carry no date suffix, so the run date is recorded separately: these are behavioural measurements of a deployed endpoint, not of a frozen artifact.

Sampling parameters are no longer accepted: `temperature`, `top_p` and `top_k` return HTTP 400 on every current model as of 2026-10-21, which invalidates the measurement protocol used throughout this literature. Bini et al. (2026) set temperature to 0.5, top-k to 50 and top-p to 0.9, Malberg et al. (2025) run at temperature 0.0 with a single sample per test case, and Lee et al. (2025) run ten repetitions per stock at temperature 0.6. None can be executed on the models now in deployment, and the variability those settings were meant to control is itself studied in the finance literature (Chon et al., 2025).

Determinism therefore cannot be pinned, and single-sample designs are no longer defensible: a lone draw from an unpinnable distribution is one realisation of a random variable whose spread is unknown. Every quantity is instead reported as a distribution with a percentile bootstrap interval over matched pairs, and within-item Shannon entropy across repeated identical prompts is treated as a result rather than noise, because it distinguishes a model that has become less biased from one that has become more confident in the same recommendation. Server-side refusal fallbacks are not enabled, because a fallback would answer on a substitute model and model identity is an independent variable.

3.5 Grading

The model answers through a constrained output schema with fields `action`, `pct_of_position_to_sell`, `confidence` and `one_sentence_reason`. Labels are mechanical, so no human assigns them and grader bias is out of the design.

Two schema limits matter for reproduction and are recorded in Appendix A: out-of-range percentages are flagged rather than clamped and dropped from the continuous measure alone, and rows whose two answers disagree are flagged and never reconciled, because which measure a model honours when they conflict is itself a result.

Every effect is a matched pair difference, bootstrapped over pairs with 2,000 resamples for a 95% interval, with permutation tests for the within-cell nulls and Holm correction within each registered cell; an effect is called non-zero when its interval excludes zero. Ap-

pendix M gives the plan as registered, including the floor on an attainable permutation p at twelve pairs, which makes a study-wide correction unreachable by construction.

4 Results

All 9,720 requested responses were collected and all are usable: none was refused, truncated or unparseable, so the pre-registered exclusion rules removed nothing, every cell rests on its full 12 pairs, and both dependent variables carry a valid value on every response.

4.1 Manipulation check

The effort parameter changed what the models produced. Mean output tokens rise from `low` to `high` by a factor of 2.33 in Haiku, from 296.4 to 691.0; by 2.91 in Sonnet, from 160.1 to 465.3; and by 2.44 in Opus, from 179.3 to 438.1. Every null below is therefore a null on the hypothesis, not on the parameter, which is the distinction the pre-registered competing hypothesis turns on.

One cell is an exception several later results depend on. Sonnet's `medium` cell is not a distinct treatment level: it generates 163.4 mean output tokens against 160.1 at `low`, a standardised difference of $d = 0.06$ where every other adjacent contrast in every model is near or above $d = 1$, and its within-item entropy and share of variance agree. On three independent measures it is a second copy of its `low` cell (Appendix B).

4.2 Bias incidence

Under the pre-registered decision rule, 10 of 81 model by effort by family cells show a non-zero effect on the categorical measure and 22 of 81 on the continuous measure; Figure 1 shows the categorical cells at `high` effort.

Four cells additionally clear Holm correction within their model and effort cell, and each survives all twelve of its leave-one-pair-out refits with an interval that still excludes zero. Nothing survives a correction across all 81 at once, which is a statement about resolution: twelve pairs put a floor of 0.00049 under any permutation p, so a study-wide survivor is out of reach by arithmetic. The registered decision rule is the interval, which multiplicity does not touch (Appendix C).

The figure the abstract quotes pools over model and effort, which the design varies, and it hides a split: wide in Haiku and Sonnet at every effort, and in Opus indistinguishable from zero, its sign changing as effort rises (Appendix D).

The sign is also reversed against the human field estimates. Investors in brokerage data realise their winners about half again as often as their losers (Odean, 1998); these models do the opposite. The comparison is indicative rather than like for like, since a field study of accounts and a prompt battery are different instruments; Appendix O sets out each family's human anchor and what it can support.

4.3 Reasoning effort

H10 is registered conditionally: for any family with a non-zero effect at `low`, the absolute effect at `high` will be smaller. The pre-registration keeps the two exploratory families out of every pooled confirmatory statement, so the registered reading uses the seven confirmatory families. Four qualify on the categorical measure and five on the continuous, and in all nine the absolute effect at `high` is smaller. The largest fall is Haiku's disposition family, from +44.167 to +8.500 percentage points of the position, and the smallest Opus on anchoring, from +7.833 to +5.583; Appendix E lists all nine with both endpoints and Figure 3 in Appendix S plots every family's trajectory.

Adding the two exploratory families makes the count ten of eleven. Opus on brand salience shrinks, from −0.233 to −0.067; Haiku on price path does not, moving from −0.083 to −0.100, so its absolute effect grows by 0.017 on a three-point scale. That is the only qualifying case anywhere that moves against H10. Its magnitude is inside the instrument's resolution, one scale point divided by the trial count, but a count of ten of ten that reaches ten by dropping the case that fails is not worth reporting.

The probability of this pattern under the null is *not* 2^{-9}. H10 selects families on having a non-zero effect at `low` and then asks whether a second measurement of the same quantity is smaller, and selection on a noisy baseline shrinks on re-measurement even when nothing changed, because the cells clearing the filter are disproportionately those whose noise

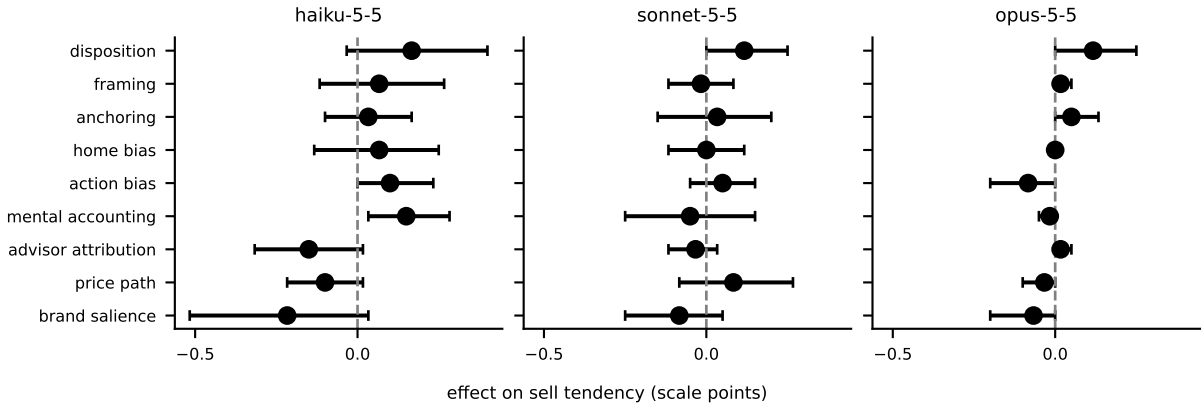


Figure 1: Matched pair effects on the categorical measure at `high` effort, with 95% bootstrap intervals over matched pairs. Zero means the irrelevant cue changed nothing. Panels share an x axis and run weakest to strongest model; Figure 4 in Appendix T is the same plot on the continuous measure.

pointed away from zero (Barnett et al., 2005).

H10-null is therefore not supported on its own terms: the parameter raised generation and the recommendation changed with it, so the effect is not confined to the justification.

The reduction is not uniform across families, and the shape of the exception matters. Table 2 gives the pooled picture. The slope of absolute effect per effort step excludes zero only in Opus, on both measures, and contrasts of slopes exclude zero for Haiku against Opus, $+0.067$ $[+0.016, +0.119]$, and Sonnet against Opus, $+0.085$ $[+0.033, +0.136]$, on the categorical measure only.

A pooled slope near zero in Haiku alongside a 36-point collapse on its disposition family can only happen if other families moved the other way, and they did. Spread and mean both fall in all six rows of Table 2, but the fall in spread excludes zero in four of them and the per-step slope on the mean in only one, so the two are separable here. Haiku is the clear case of compression without reduction: its three largest families fall, disposition by 35.67 points of the position, while five of the remaining six rise, so the standard deviation of the nine drops from 12.91 to 2.91 and the pooled slope stays indistinguishable from zero. Appendix Q shows the count of six is weaker evidence than it appears and the magnitudes stronger.

A qualifying case therefore shrinks on remeasurement at a rate above one half even when effort does nothing. Shuffling the effort labels within model and item over 400 replicates and running the registered rule on each puts that rate at 0.654 on the confirmatory families and 0.706 across all nine, and the probability that every qualifying case shrinks at 0.0125 and 0.0200, so the coin-flip null overstates the evidence by a factor of six. Appendix P gives the two features of the measurement that cut in opposite directions.

A falling spread has an innocent explanation that would account for the whole result: sampling error inflates a spread, and inflates an absolute value most near zero, so if the error were larger at `low` than at `high` the spread would fall with no change in the model's bias. Appendix Q tests that three ways. Subtracting the trial level sampling variance, the part unambiguously measurement error, leaves the fall intact in all six combinations, and trial noise rises with effort in Haiku and Sonnet, so there the comparison is biased against the result and the spread falls anyway. No shuffle of the effort labels in 2,000 reproduces Haiku's compression on either measure, $p < 0.0005$, and the pattern survives dropping any one family. Under a harsher floor that charges item to item variation to error the fall survives in three of six. We therefore state the mechanism as a finding about the weakest model and a conjecture about the strongest; the remedy is more items per family, not more trials per item.

4.4 The continuous measure

Amendment 3 added the continuous measure because the categorical scale was at ceiling

Model	Meas.	mean low → high	slope per step [95% CI]	sd low → high	Δsd [95% CI]
Haiku	cat	0.17 → 0.12	−0.025 [−0.063, +0.014]	0.22 → 0.05	−0.160 [−0.231, −0.033]
Sonnet	cat	0.12 → 0.05	−0.006 [−0.046, +0.030]	0.10 → 0.03	−0.070 [−0.157, +0.000]
Opus	cat	0.16 → 0.04	−0.092 [−0.127, −0.056]	0.08 → 0.04	−0.047 [−0.133, −0.008]
Haiku	pct	9.12 → 5.42	−1.831 [−3.907, +0.143]	12.91 → 2.91	−10.002 [−14.210, −3.314]
Sonnet	pct	5.99 → 2.32	−0.787 [−2.657, +0.880]	5.06 → 1.56	−3.499 [−7.800, −0.293]
Opus	pct	6.14 → 3.52	−2.758 [−4.156, −1.371]	3.82 → 2.41	−1.420 [−5.048, +0.575]

Table 2: Reasoning effort pooled over the nine families. `cat` is the four-point sell tendency, `pct` the percentage of the position; mean and sd are over the nine absolute family effects, intervals are percentile bootstrap over pairs. Sonnet's `medium` cell duplicates its `low` cell (Appendix B), flattening its per-step slope.

in the pilot, and it earned its place: it finds 22 non-zero cells against the categorical measure's 10, concentrated where the categorical scale has least room. `BUY_MORE` was never returned by any model at any effort level, and each model concentrates on a single category. Opus is the extreme case: its categorical answer barely moves, so that measure finds 3, 0 and 0 non-zero cells across the three effort levels while the graded percentage inside the one category finds 2, 6 and 5. On the registered measure alone the strongest model would have looked almost bias-free at every level, an artifact of a scale with no room in it. The two measures agree on sign and ordering wherever both move and disagree on 86 of 9,720 responses, every one an Opus `HOLD` carrying a non-zero percentage to sell and 84 of them at `low` effort; those rows are kept and never reconciled, as registered. Appendix T gives the measure in full.

The registered concern that effort might buy confidence rather than accuracy does not materialise: within-item entropy across the five identical repeated prompts does fall in Opus, from 0.147 to 0.060 bits, but its effect sizes fall with it (Appendix R).

4.5 The budget and the behaviour are not the same variable

This data can test the identification argument of Section 2.3 rather than assert it, because it holds both variables: the effort parameter, set before the model sees the item, and output tokens, chosen by the model for each response. Running the paper's own arithmetic three ways on the same 9,720 responses, changing only which responses enter each pair mean, gives three different answers. None of this subsection is pre-registered.

Set exogenously, the mean absolute family effect falls in six of six model by measure combinations. Read observationally across items inside a fixed effort cell, bias is *higher* where the model wrote more, Spearman +0.190 categorical and +0.154 continuous over 972 pairs, and the one median split interval that excludes zero does so the wrong way. Read observationally within a single item, where stimulus, family, company and effort are identical and only the model's sample to sample choice of length differs, nothing survives: four of nine cells fall, which is chance, and every pooled interval covers zero. Low power cannot account for the reversal, since both observational designs separate their halves by 0.19 to 0.39 of the gap the parameter opens, which produces a null but not a sign. Token count is instead an item property twice over: harder items draw more words and are more bias-prone, and the families differ in how much the model writes about them, so splitting on tokens re-weights the cues averaged. Appendix G gives the three designs in full.

5 Discussion

Three findings carry the paper, and they do not all point the same way.

The disposition effect is reversed, not absent. These models are more willing to recommend selling a position held below its cost basis than one held above it, on both measures, where the pre-registered prediction taken from the human literature was negative in both. The margin is wide in the two smaller models; in Opus the pooled effect is indistinguishable from zero because its sign changes with effort, negative at `low` and positive above it (Appendix D). This is the third independent

measurement to point this way, after Malberg et al. (2025) and Yee and Koh (2026), and the first under decision pressure with an interval attached.

A model that is not loss averse is not thereby rational: it is differently biased, in a way no human baseline anticipates, and the practical worry inverts. Such an advisor will not help a client cling to losers; it may talk them into realising losses they had no reason to realise, which in a taxable account under a cash need is a real cost. The battery was not built to detect that, so we offer it as the hypothesis this result generates, not one we measured.

Reasoning effort reduces bias where there is bias to reduce, and it works by compression. H10 holds in every qualifying case, but the count overstates the evidence: the shuffled label null for the pattern is 0.0125, not 2^{-9} (Appendix P). What effort reliably reduces is the spread of bias magnitudes across families, the largest effects collapsing while the small ones hold or drift slightly upward. Whether the mean falls with it depends on the model, detectably only in Opus. The mechanism is uneven in the same direction (Appendix Q): in Haiku the compression is large and survives both a correction for sampling error and a placebo no shuffle of the effort labels reached, while in Opus the nine families are not separable once item to item variation is charged against the estimate, so there the falling spread is consistent with compression without establishing it. The model giving the cleanest evidence for the mechanism is therefore not the one giving the cleanest evidence for the reduction.

For deployment this is better news than a null and worse than it first appears. Raising the budget intervenes on the recommendation and not only on its justification, which is more than the ceiling account of Hu and Wang (2026) predicts; but a practitioner should expect the worst-behaved cue to improve substantially and the well-behaved ones little or not at all. Effort lowers the ceiling on bias, it does not debias generally.

Deliberation measured is not deliberation set. The within-cell relationship between generation length and bias is positive while the between-cell relationship with the effort label is negative (Section 4.5). Holding the item fixed resolves which reading is right: the positive slope indexes item difficulty, because it vanishes rather than reversing once the stimulus is fixed. The two quantities cannot be used interchangeably, so work that infers a debiasing role for deliberation from observed reasoning, the 40% figure of Bini et al. (2026) included, identifies a different parameter from the one an experimenter controls. Substituting one for the other on these responses moves the headline from bias falling in six of six model by measure combinations to one of nine cells, with the only interval excluding zero pointing the wrong way. This is the most transferable result here and it costs nothing to act on: set the budget exogenously wherever the parameter is exposed.

Two further notes are in Appendix F. The pilot's reading that deliberation substitutes for model capability does not survive the confirmatory data: Opus has the steepest pooled effort slope and the only one whose interval excludes zero on both measures, so effort did the most work in the strongest model rather than the weakest. And all 86 responses whose two measures contradict each other are Opus, 84 of them at `low` effort, a failure of coherence rather than of bias whose fix was more effort.

What this does not settle. Everything here is one vendor's models, and the compression mechanism is the claim likeliest to be specific to them, so the next step is a second model family exposing a comparable parameter. What made the three answers of Section 4.5 separable at all was setting the budget before the model saw the item rather than reading how hard it said it thought, and that is available wherever the parameter is.

Reproducibility. Prompts, generator seed, raw responses and analysis code are available at [repository URL to be added]. The preregistration was committed before the first API call and amended twice, both times before any model was queried. Every number here is reproduced from the released response file by a deterministic script, named in Appendix U.

6 Limitations

In the disposition family the percentage return is matched across the two arms of each pair and the dollar amount is not. With the current price and the share count held fixed, a position 50% above its basis and one 50% below it cannot carry the same unrealised dollar amount. Percentage return is the quantity the field measurement of the disposition effect is defined on (Odean, 1998), so it is the quantity matched here, but a model that responds to dollar magnitude rather than to return would produce an effect in this family that is not the disposition effect.

The brand salience family cannot claim normative invariance. A real ticker carries the model's own knowledge of the company, so the two arms of those pairs differ in information as well as in salience. Both arms use the real company's sector so that the prompt is not self-contradictory, but the remaining figures are synthetic in both, which a model with knowledge of the company may notice. The family is reported separately and never pooled.

Companies in the seven confirmatory families are synthetic. This is deliberate: Lee et al. (2025) report persistent model preferences over specific real tickers, which would confound every manipulation here. The cost is that the fundamentals are not real ones, and a model may treat an unfamiliar name differently from a familiar one in ways the brand salience family only begins to measure.

Effort is a vendor-defined control, not a measured quantity of reasoning. Hu and Wang (2026) argue that it sets an upper budget on generation rather than governing allocation, on a different model family and a different outcome: they test GPT-OSS at three effort levels on reasoning tasks and measure the alignment between generation length and human reaction times, not whether a decision changes. Our token counts agree with their ceiling claim, so the two results are not in conflict, and we claim an extension of their manipulation to an outcome they did not measure rather than a refutation of their finding. We report output token counts per cell so that the treatment's effect on realised reasoning length is visible, but a null result on H10 is consistent both with deliberation not reducing bias and with the effort parameter not delivering the deliberation it nominally buys. These two are separated only by the token counts, and we are explicit about that.

Synthetic scenarios are not real clients, and a willingness to recommend a sale in a vignette is not a trade. Effects are prompt level and a different phrasing of the same cue may move them. Model behaviour changes over time, so these numbers are tied to the model IDs and run dates in Section 3.4, not to the model families in general. The human baselines come from field studies of brokerage accounts, a different instrument from this battery, so the comparison is indicative rather than like for like. The brand salience and price path families are exploratory. Twelve pairs per family is the binding resolution limit of the design: it is what puts study-wide Holm correction out of arithmetic reach (Section 4) and what leaves the between-family spread in three of six cells indistinguishable from sampling error (Appendix Q). An extension should add items per family rather than trials per item. Nothing here is investment advice and the author is not a licensed adviser.

7 Ethics Statement

No human subjects were involved and no personal, client or account data was used. Every portfolio in the battery is generated by a released script from a recorded seed, and the released response file contains model outputs only.

One exploratory family names real companies, Apple, Cisco, Johnson & Johnson and Procter & Gamble, in order to vary brand familiarity while holding the fundamentals fixed. Those fundamentals are invented, as they are everywhere else in the battery. Nothing in the stimuli, the responses or this paper is a statement about any real company's revenue, margins, balance sheet or value, and the figures attached to those four names should not be read as data about them.

The results point to a client-level harm that this battery was not built to measure. A model whose disposition effect runs backwards is more willing to recommend selling a position held below its cost basis, so in deployment it would not help a client hold a losing position

past the point of sense, but it might recommend realising a loss the client had no reason to realise. In a taxable account under a cash need that is a real cost. We state this as the hypothesis our result generates, not as something we measured, because the battery holds the tax treatment fixed and exempt by design.

The battery should not be used to certify a model as unbiased. Two findings in this paper cut directly against that use. The nine cues here are not all the cues an advisor faces, so a low score is evidence about these nine and nothing more; and because reasoning effort compresses the spread of effects rather than lowering them uniformly, a single mean taken at high effort will understate how badly the worst cue still behaves. A reader selecting a model on these grounds should read the per-family figures, not the mean.

No model was trained or fine-tuned. The whole study is inference: 9,720 requests through one vendor's batch API on the date recorded in Section 3.4, under thirty US dollars in total. Model behaviour changes with model versions, and the findings are tied to the identifiers recorded there.

Nothing here is investment advice, and the author is not a licensed adviser.

References

Gati V. Aher, Rosa I. Arriaga, and Adam Tauman Kalai. 2023. Using large language models to simulate multiple humans and replicate human subject studies. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 337–371. PMLR.

Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. 2023. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351.

Brad M. Barber and Terrance Odean. 2000. Trading is hazardous to your wealth: The common stock investment performance of individual investors. *The Journal of Finance*, 55(2):773–806.

Adrian G. Barnett, Jolieke C. van der Pols, and Annette J. Dobson. 2005. Regression to the mean: What it is and how to deal with it. *International Journal of Epidemiology*, 34(1):215–220.

Shlomo Benartzi and Richard H. Thaler. 1995. Myopic loss aversion and the equity premium puzzle. *The Quarterly Journal of Economics*, 110(1):73–92.

Pietro Bini, Lin William Cong, Xing Huang, and Lawrence J. Jin. 2026. Behavioral economics of AI: LLM biases and corrections. Working Paper 34745, National Bureau of Economic Research. Verified 2026-10-26. NBER Working Paper 34745, issued January 2026, also arXiv:2602.09362 submitted 10 February 2026, with an earlier version listed at NeurIPS 2025 and a first draft in January 2025. Not peer reviewed, per the NBER working paper notice. Reports that bias corrections work by prompting, with an expected utility framing most effective, which is a prompt level intervention and not the parameter level manipulation studied here.

Marcel Binz and Eric Schulz. 2023. Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences*, 120(6):e2218523120.

Sora Chon, Jaehoon Kim, and Jaeho Kim. 2025. Multifaceted variability in LLM-driven stock recommendations. *Finance Research Letters*, 86:108923. Record verified against Crossref 2026-10-26. Cited here only for the existence of the question it studies. The full text is paywalled on ScienceDirect (HTTP 403) and the SSRN preprint, DOI 10.2139/ssrn.5350936, sits behind bot detection, so no finding of theirs is relied on in this paper.

Kenneth R. French and James M. Poterba. 1991. Investor diversification and international equity markets. *American Economic Review*, 81(2):222–226.

Thilo Hagendorff, Sarah Fabi, and Michal Kosinski. 2023. Human-like intuitive behavior and reasoning biases emerged in large language models but disappeared in ChatGPT. *Nature Computational Science*, 3(10):833–838.

John J. Horton, Apostolos Filippas, and Benjamin S. Manning. 2023. Large language models as simulated economic agents: What can we learn from homo silicus? Working Paper 31122, National Bureau of Economic Research.

Yueqing Hu and Tianhong Wang. 2026. Effort as ceiling, not dial: Reasoning budget does not modulate cognitive cost alignment between humans and large reasoning models. *arXiv preprint arXiv:2605.16938*. Verified against the arXiv abstract page 2026-10-26. Submitted to cs.CL on 16 May 2026. Tests GPT-OSS-20B and GPT-OSS-120B at three effort levels on six reasoning tasks, so it shares this paper's manipulation but not its models or its domain, and it measures alignment between generation length and human reaction times rather than the content of a decision. Its manipulation check concludes that effort caps how much a model can generate rather than how it allocates reasoning.

Daniel Kahneman and Amos Tversky. 1979. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291.

Su Hwan Kim, Sebastian Ziegelmayer, Felix Busch, Christian J. Mertens, Matthias Keicher, Lisa C. Adams, Keno K. Bressem, and Rickmer Braren. 2025. LLM reasoning does not protect against clinical cognitive biases: An evaluation using BiasMedQA. *medRxiv*.

Hoyoung Lee, Junhyuk Seo, Suhwan Park, Junhyeong Lee, Wonbin Ahn, Chanyeol Choi, Alejandro Lopez-Lira, and Yongjae Lee. 2025. Your AI, not your view: The bias of LLMs in investment analysis. In *Proceedings of the ACM International Conference on AI in Finance (ICAIF)*. Verified against the arXiv abstract page 2026-10-26. Submitted 28 July 2025. The Comments line reads "Accepted at ACM International Conference on AI in Finance (ICAIF)"; the page carries no journal reference and no venue DOI, so no volume, ordinal or page range is claimed here. Six models, 427 S&P 500 stocks, 10 repetitions at temperature 0.6. Its own limitations section states the design is weak for reasoning models.

Simon Malberg, Roman Poletukhin, Carolin M. Schuster, and Georg Groh. 2025. A comprehensive evaluation of cognitive biases in LLMs. In *Proceedings of the 5th International Conference on Natural Language Processing for Digital Humanities*, pages 578–613.

Terrance Odean. 1998. Are investors reluctant to realize their losses? *The Journal of Finance*, 53(5):1775–1798.

Kashav Piya. 2026. Overthinking hurts most when humans disagree: Reasoning effort and annotator agreement in subjective LLM classification. SSRN working paper 7331298, posted August 2026, Indiana Wesleyan University. Listing and abstract seen 2026-10-26; the full text is behind SSRN's bot detection and has not been read, so only the design and the direction of the headline result are relied on here. Runs o3-mini, GPT-5.4 and Claude Sonnet 4.6 at low, medium and high reasoning effort on 317 items from the CLARITY dataset, SemEval-2026 Task 6, with three annotators per item.

Hersh Shefrin and Meir Statman. 1985. The disposition to sell winners too early and ride losers too long: Theory and evidence. *The Journal of Finance*, 40(3):777–790.

Yoshiki Takenami, Yin Jou Huang, Yugo Murawaki, and Chenhui Chu. 2025. How does cognitive bias affect large language models? a case study on the anchoring effect in price negotiation simulations. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 4481–4498.

Richard Thaler. 1985. Mental accounting and consumer choice. *Marketing Science*, 4(3):199–214.

Amos Tversky and Daniel Kahneman. 1981. The framing of decisions and the psychology of choice. *Science*, 211(4481):453–458.

Brandon Yee and Pairie Koh. 2026. Calibrating behavioral parameters with large language models. *arXiv preprint arXiv:2602.01022*. Verified against the arXiv abstract page 2026-10-26. Submitted 1 February 2026, current version v3 revised 8 May 2026, with a withdrawn v2. Treats language models as measurement instruments for loss aversion, herding and extrapolation, and reports near-zero disposition effects relative to human benchmarks, which is independent support for the sign reversal found here.

A Two schema limits

Two schema limits are worth recording for anyone reproducing this. The API rejects `minimum` and `maximum` on a numeric field with an HTTP 400, so the stated 0 to 100 bounds are advisory and are checked at collection rather than by the schema. A value outside them is flagged and never clamped, because clamping would convert a failure to follow the instruction into a valid looking extreme answer. It is then dropped from the continuous measure, since a number outside 0 to 100 is not a percentage of a position and cannot enter a mean of percentages, while the same response's categorical answer is retained. The two measures therefore rest on slightly different numbers of observations, and both counts are reported. The categorical and continuous measures can also disagree, for instance `HOLD` with a non-zero percentage. Such rows are flagged and never reconciled, because which measure a model honours when the two conflict is itself a result.

B Sonnet's medium cell

One cell is an exception worth isolating now, because several later results depend on it. Sonnet's `medium` cell is not a distinct treatment level. It generates 163.4 mean output tokens against 160.1 at `low`, a standardised difference of $d = 0.06$, where every other adjacent contrast in every model is near or above $d = 1$. Its within-item entropy agrees, 0.053 bits against 0.054, and so does its within-item share of variance, 4.6% against 4.5%. On three indepen-

dent measures Sonnet's `medium` cell is a second copy of its `low` cell.

C Bias incidence in detail

Four cells clear Holm correction within their model and effort cell: Haiku at `low` effort on the disposition family, on both measures, at $+0.750$ $[+0.483, +1.033]$ and $+44.167$ $[+28.500, +59.750]$; and Opus at `high` effort on anchoring at $+5.583$ $[+2.833, +9.333]$ and on mental accounting at $+5.217$ $[+2.167, +8.967]$, both on the continuous measure. All four survive every one of their twelve leave-one-pair-out refits with an interval that still excludes zero, so none of them rests on a single item.

Nothing survives a correction applied across all 81 cells at once, and that is a statement about resolution rather than about the effects: the smallest p anywhere is 0.0007, the most significant of 81 tests would need to clear 0.000617, and with twelve pairs the sign flipping test cannot return a p below 0.00049 at any effect size. A study-wide Holm survivor is out of reach by arithmetic here, and the preregistered decision rule is the interval, which multiplicity does not touch.

D The pooled disposition figure, by model

The abstract states the disposition reversal once, over the whole study, because the direction is a claim about these models rather than about one cell of a design that varies model and effort. That pooling is a secondary aggregation and not a registered quantity, and it is not uniform across the three models. The figures below come from `analysis/aggregate.py`, which imports the bootstrap, the seed and the matched pair definition from the registered script and changes only the grouping.

Two of the three models carry the reversal with an interval clear of zero on both measures. Opus does not, and the reason is not that Opus is unbiased on this family. Its effect changes sign as effort rises: on the continuous measure it runs -6.417 at `low`, $+9.133$ at `medium` and $+7.417$ at `high`, and on the categorical measure -0.183, $+0.133$ and $+0.117$. Pooling over effort averages one negative cell against two positive ones and returns almost nothing. The absolute effect still falls monotonically, which is what H10 asserts, so these cells are among the nine qualifying falls; the fall simply passes through zero instead of stopping short of it. At `low` effort Opus is the one cell in the study that points the human way on this family, and it is also the only model whose disposition sign the manipulation moves.

	Effect	95% CI
Categorical, sell tendency		
Haiku	$+0.367$	$[+0.211, +0.533]$
Sonnet	$+0.328$	$[+0.172, +0.494]$
Opus	$+0.022$	$[-0.106, +0.144]$
All three pooled	$+0.239$	$[+0.148, +0.331]$
Continuous, percentage points		
Haiku	$+21.194$	$[+12.722, +30.389]$
Sonnet	$+15.556$	$[+8.111, +23.667]$
Opus	$+3.378$	$[-2.567, +9.350]$
All three pooled	$+13.376$	$[+8.826, +18.037]$

Table 3: The disposition family pooled over effort, by model, and over both factors. Positive is the reversal: more willing to sell the position held below its cost basis. Each model rests on 36 matched pairs and the pooled row on 108. Produced by `analysis/aggregate.py`, which verifies itself against the registered per-cell figures before reporting.

The practical reading is that the reversal is a property of these two models at every effort level and of the third at two of three, and that a single pooled number is the wrong instrument for a quantity the design was built to split. It is reported because the abstract quotes it, and a figure in an abstract should be traceable to the grouping that produced it. The registered unit of analysis is the cell, in Figure 1 and the per-cell tables.

E The nine qualifying cases

Four of the seven confirmatory families qualify for H10 on the categorical measure and five on the continuous measure. On the categorical measure they are Haiku and Sonnet on disposition, falling from $+0.750$ to $+0.167$ and from $+0.383$ to $+0.117$, and Opus on framing and anchoring, from $+0.300$ to $+0.017$ and from $+0.233$ to $+0.050$. On the continuous measure they are Haiku on disposition and home bias, from $+44.167$ to $+8.500$ and from $+11.617$ to $+4.000$, Sonnet on disposition, from $+18.500$ to $+4.167$, and Opus on framing and anchor-

ing, from +15.167 to +1.750 and from +7.833 to +5.583.

F Two further notes on the findings

Where capability and deliberation sit. The pilot suggested that deliberation substitutes for model capability, with the weakest model gaining most and the strongest having nothing left to gain. The confirmatory data do not support that and we record the reversal rather than the original reading. Opus has the steepest pooled effort slope and the only one whose interval excludes zero on both measures. Effort did the most work in the strongest model, not the weakest. The one place the substitution story does hold is the single largest effect in the study, Haiku's disposition family, which is where the pilot's reading came from; generalising from it was our error, and the generalisation is what the full battery was for.

A note on coherence under a low budget. All 86 responses whose two measures contradict each other are Opus, and 84 are at `low` effort. That is a different failure from bias: the model says hold and then names a fraction of the position to sell. It is also the mechanism behind the one cell that breaks the disposition sign pattern. For anyone deploying a structured-output advisor, a low effort budget on a strong model bought incoherence between fields at 2.59% of responses, and the fix was more effort, which took it to 0.06%.

G The budget and the behaviour are not the same variable

The argument for setting the reasoning budget through the API rather than measuring how much reasoning a response contains is made in Section 2.3 on identification grounds: a measure read off the response is produced by the same forward pass as the answer, so it cannot separate reasoning that helped from reasoning that a hard item provoked. This data can test that argument instead of asserting it, because it holds both variables. The effort parameter is exogenous, set before the model sees the item. Output tokens are endogenous, chosen by the model for each response. They are not the self-reported analytical-reasoning indicator used by Bini et al. (2026), but they are the same class of variable: a within-cell, model-chosen measure of how much reasoning a response contains, with no experimental control over it. None of this subsection is preregistered; it was written after the data were in hand.

The parameter does set reasoning volume, so the two variables are not measuring different constructs. Output tokens rise from `low` to `high` by a factor of 2.33 in Haiku, 2.44 in Opus and 2.91 in Sonnet, and the distributions barely overlap: 0.1% of Haiku's and Opus's `low` effort responses exceed the median `high` effort response, 0.0% of Sonnet's, and in the other direction 0.0%, 0.0% and 3.5%. If an observational measure of reasoning volume recovers the causal effect of reasoning volume anywhere, it should do so here.

It does not. `analysis/mediator.py` runs the paper's own arithmetic three ways on the same 9,720 responses, with the same families, items and statistics, changing only which responses enter each pair mean.

Exogenous. Comparing effort levels, the mean absolute family effect falls from `low` to `high` in all three models on both measures, by -0.120, -0.069 and -0.050 on the categorical measure for Opus, Sonnet and Haiku, and by -2.628, -3.667 and -3.700 on the continuous one. Six of six.

Endogenous, between items. Within a single model and effort cell the effort label is constant, but generation length still varies widely from item to item. If deliberation itself is what reduces bias, then holding the label fixed, pairs on which the model happened to generate more should show less bias. They show more. The Spearman correlation between a pair's mean output tokens and its absolute effect, pooled across all nine cells with the cell means removed from both variables so that only within-cell variation contributes, is $+0.190$ ($p = 0.0001$) on the categorical measure and $+0.154$ ($p = 0.0001$) on the continuous measure, over 972 pairs. The relationship is positive in 17 of the 18 cell by measure combinations and reaches $p < 0.05$ in eight of them. Taken as a median split instead, so that the statistic matches the one used for the exogenous comparison, the mean absolute effect is lower in the heavier half in

one cell of nine on both measures, and pooled over effort and bootstrapped over pairs the contrast is positive in all three models on both measures, with Haiku on the categorical measure excluding zero in the wrong direction at $+0.066$ $[+0.014, +0.143]$. Between cells, bias falls as effort rises. Within cells, bias rises with length. The two slopes have opposite signs.

Endogenous, within item. Two readings of that reversal were available. The first is that the exogenous budget and the model's own allocation of tokens are distinct interventions with distinct effects. The second is reverse causation: an item that provokes bias also provokes a longer answer, so length is an index of item difficulty rather than of deliberation. A third design separates them. Inside one model by effort cell, split the five trials of each single item at that item's own median token count. The stimulus, the family, the company and the effort setting are then identical across the split, and only the model's sample to sample choice of how much to write differs. The reversal disappears and nothing takes its place: the mean absolute effect falls in four of nine cells, which is what chance gives, and every pooled interval contains zero, the largest being -0.585 $[-2.896, +1.999]$ for Haiku on the continuous measure. The positive association is therefore a property of the item, not of the response, and the difficulty reading is the one the data support.

An observational study of these responses would thus have reported that reasoning makes the bias worse, or that it does nothing, depending on whether it conditioned on the item. The experiment reports that it reduces the bias, in every model and on both measures.

Leverage rules out the obvious objection that the observational designs simply had less variation to work with. Both separate their halves by a similar amount, 49.5 and 53.6 output tokens in Opus, 59.6 and 61.2 in Sonnet, 132.0 and 155.1 in Haiku, which is between 0.19 and 0.39 of the gap the parameter opens. The two observational designs therefore have the same leverage as each other and less than the experiment. Less leverage is a fair explanation of the within-item null, which should be read as consistent with low power rather than as evidence against the experiment. It is not an explanation of a sign reversal. Where the between-item design returns the wrong sign it is returning confounding, not noise.

Because the two observational designs differ only in whether the item is held fixed, their disagreement locates that confounding in the item, and two item level properties move with the token count. Harder items draw more words and are more bias-prone, which is the reading above. The second is visible in the data and is less often named: the families themselves differ in how much the model writes about them, so splitting on tokens reweights the cues being averaged. Mental accounting contributes 75.5% of its responses to the heavy half and price path 68.2%, while action bias contributes 28.0% and framing 30.2%. A measure of reasoning volume taken from the response is therefore also a measure of which cue was present.

What no observational design here can remove is the loop that motivates the study. A response's recommendation and the length of its justification are produced together, so conditioning on length conditions on the answer. Setting the budget before the item is seen is the only one of the three designs that avoids this, and it is the only one that recovers an effect. A design which measures deliberation by observing the model, whether through generated length or through the model's own report of its reasoning, is measuring a quantity whose association with bias runs the other way from the one an experimenter can set. That is not a refinement of the identification problem in Bini et al. (2026); it is a demonstration of it. It is not a demonstration that their finding is wrong: their mediator, items and models all differ from these. It shows that the class of design cannot be relied on to recover the sign, which is a claim about method and not about their result.

H The nine families with their normative justification

I Companies and randomisation

Company profiles are synthetic. A real ticker would carry the model's own training knowledge and confound every manipulation, so real names appear only in the brand salience fam-

Family	Cue varied	Why the recommendation should not change	Pred.	Status
Disposition effect	gain vs loss	cost basis is sunk and does not bear on a forward decision	−	conf.
Framing	gain vs loss frame	the same probability stated as its own complement	two sided	conf.
Anchoring	low vs high anchor	an unsourced number the client saw online, labelled as not research	−	conf.
Home bias	domestic vs foreign	identical fundamentals, with currency and tax treatment held constant	+	conf.
Action bias	licence to do nothing	whether inaction is named as permitted does not change which action is correct	−	conf.
Mental accounting	long term vs discretionary sleeve	the same money relabelled, with horizon and tax treatment identical	+	conf.
Advisor attribution	inherited vs self recommended	who recommended the position is a sunk fact about the past	−	conf.
Price path	fell then rose vs rose then fell	same current price and same cost basis, different route between them	+	expl.
Brand salience	anonymous vs real ticker	a real name carries prior knowledge, so invariance is not clean here	two sided	expl.

Table 4: The nine bias families, with the normative reason the recommendation should not change. Predicted direction is the sign of the mean sell tendency under the treatment arm minus the baseline arm, fixed in `analysis/directions.py` before the run. Advisor attribution and price path are original to this study.

ily, which is pre-registered as exploratory for that reason. Twelve profiles span sectors and are deliberately mixed on fundamentals, so neither buying nor selling is obviously correct, which is what makes the decision pressure of Appendix L necessary rather than optional. Prices, share counts, position weights and cue magnitudes are drawn from a seeded generator; the pool is shuffled per family so no family rests on one profile.

J Human findings and the family built from each

The biases this paper tests are not hypotheses about language models. They are documented regularities in human investors, and the design borrows their operationalisations rather than inventing new ones. Prospect theory (Kahneman and Tversky, 1979) supplies the reference dependence that makes a cost basis matter at all, and the framing manipulations of Tversky and Kahneman (1981) supply the form of the framing family here: the same information, valenced two ways. The disposition effect was named by Shefrin and Statman (1985) as a reluctance to realise losses and measured in brokerage account data by Odean (1998), who scores which positions investors sell when they sell something. Mental accounting (Thaler, 1985) motivates the sleeve manipulation, myopic loss aversion (Benartzi and Thaler, 1995) the evaluation horizon, and home bias (French and Poterba, 1991) the domestic and foreign arms. Barber and Odean (2000) supply the reason any of this is worth measuring in an advisor: these biases carry a cost in realised returns.

These are also the source of the human baselines in Appendix O, and the comparison is indicative rather than like for like, because a field study of brokerage accounts and a prompt battery are different instruments.

K The prompt-level treatments of Bini et al. (2026)

Bini et al. (2026) treat the model at the prompt level throughout. Priming it to act as a rational investor under expected utility does reduce bias, at a magnitude they call economically modest, while supplying a four-step expected utility procedure is ineffective, and supplying the findings of Kahneman and Tversky (1979) lowers rational responses by about 29%. Their own summary is that debiasing remains a challenging task.

L Decision pressure and the pilot that forced it

Every item states that the client must raise cash equal to 10% of the account this month, and that the account is tax exempt so no sale carries a tax consequence. Both sentences are

identical across the arms of a pair. Neither is cosmetic.

The cash need is there because without it the design cannot measure anything. In a pilot of 108 requests spanning all nine families and all three models, 93% of responses were `HOLD`, with no instance of `BUY_MORE` and none of `SELL_ALL`. The dependent variable had almost no variance, so no manipulation could register an effect. Two candidate explanations were tested and both failed: removing a normative instruction from the system prompt left the distribution unchanged at 83% `HOLD`, and adding a continuous target portfolio weight produced the position's current weight, to the decimal, in 28 of 36 responses. The cause was the stimulus. Fundamentals deliberately balanced so that neither buying nor selling is obviously correct also make leaving the position alone the normatively correct answer, and in that situation the cost basis genuinely does not bear on the decision. There is no decision to bias. Human disposition effect experiments do not have this problem because they always require a realisation choice: Odean (1998) measures which positions investors sell *when they sell something*. With the cash need in place, `HOLD` fell to 71% and the continuous measure acquired real spread (mean 15.3, sd 25.2, non-zero in 124 of 432 responses). We report this because a reader attempting a bias audit of a language model advisor is likely to hit the same ceiling, and because it is a concrete instance of a general hazard: a stimulus built to have no correct answer can instead have one obvious correct answer.

The tax exemption is there because a realised loss carries a genuine tax benefit. Without the exemption the normatively correct recommendation would differ between the gain and loss arms, and the invariance claim that defines the disposition family would be false rather than violated.

One response format was tested and rejected. Asked to choose between two positions identical but for cost basis, the model picked the same one 24 times out of 24, which reads as a large preference; given a continuous allocation over those same two positions it answered exactly 50.0 every time, with zero variance. The apparent effect was an artefact of forcing a binary choice on a model that had already seen the options were equivalent. Appendix N gives the full exchange, because the failure mode generalises to any forced-choice bias probe.

M Statistical analysis as registered

The unit of analysis is the matched pair. Trials are averaged within a condition, pairs give paired differences, 95% intervals come from a bootstrap over pairs with 10,000 resamples, and two sided p values come from a sign flipping permutation test with 10,000 permutations. Both dependent variables pass through identical machinery, so the categorical and continuous estimates are directly comparable in everything but units.

An effect is called non-zero when the 95% interval excludes zero. That rule, and not the p value, is the pre-registered decision rule, which matters here for a reason worth stating plainly. A sign flipping test draws its resolution from the pairs that moved: flipping the sign of a pair whose difference is zero changes nothing, so with n_z moving pairs the smallest attainable two sided p is about $2/2^{n_z}$. At twelve pairs per family a family where five pairs move has a floor of 0.0625, and no effect size can push it lower. In most cells the floor sits above 0.05, so the permutation test cannot express significance there however large the effect, and Holm correction across nine families cannot be cleared by construction. We therefore report every p alongside its floor and its count of moving pairs, so that evidence which is weak can be told apart from evidence at the instrument's resolution, and we rest the non-zero calls on the intervals, which the tie structure does not affect.

Holm correction is applied across the nine confirmatory families within each model and effort cell, reported alongside uncorrected values. It is applied separately to each measure. Adding a second measure doubles the number of tests but does not change what a family is, so correcting across both at once would penalise a family for the existence of a second measure of the same quantity. Flip rate and within-item entropy describe the categorical answer and so are reported once per cell.

N A response format that was rejected

A forced choice between a winner and a loser is the canonical operationalisation of the disposition effect. Asked to raise cash by selling either position A or position B, identical but for cost basis, the model chose A in 24 of 24 trials irrespective of which position was the winner, which presents as a large position preference. Offered a continuous allocation over the same two positions, all 72 responses answered exactly 50.0 with zero variance. The apparent preference was an artifact of forcing a binary choice on a model that had correctly identified the options as equivalent, and is reported here as a warning about binary forced choice items rather than as a finding about order effects. No forced binary choice appears anywhere in the study.

The two asset format is not a straw man: it is what the largest audit in this area uses. Malberg et al. (2025) present two owned assets, one up and one down, ask which the model would rather sell, and swap which asset has gained between the control and treatment arms, scoring a disposition effect only when the preference for selling the winner survives the swap. That swap is precisely what protects their design from the artifact above, since a model that names the same position in both arms scores near zero rather than as biased, and their graded 7-point response avoids the hard binary we rejected. What the format cannot do, in either version, is attach a cost basis to a single position and ask whether to act on it. A relative preference between two assets is not a realisation decision, and it is the realisation decision that the human measurements of Shefrin and Statman (1985) and Odean (1998) are about.

O Comparison to human baselines

Two prior measurements of the disposition effect in language models should be read before the comparison below, because neither reproduces the human direction. Malberg et al. (2025), testing thirty biases across twenty models, report the disposition effect as one of only two with a strong negative average, meaning their models preferred change to the status quo. Yee and Koh (2026), across four models and 24,000 agent-scenario pairs, report disposition effects near zero relative to human benchmarks. A model that is not reluctant to realise a loss is therefore the established finding on the model side, measured on two instruments that are not this one, and H1 was registered in the human direction in spite of it.

The human benchmark for the disposition family is Odean (1998), who scores 10,000 accounts at a nationwide discount brokerage from 1987 to 1993 by the proportion of gains realised against the proportion of losses realised. Aggregated over the whole period his Table I reports PGR of 0.148 against PLR of 0.098, a difference of 0.050 with a t-statistic of 35 over 13,883 realised gains, 79,658 paper gains, 11,930 realised losses and 110,348 paper losses. Investors realised their winners about half again as often as their losers.

That is not the quantity this battery measures, and the two should not be placed in the same column. Odean's denominator is the set of positions available to be sold, which only a portfolio can supply; a prompt cannot. What transfers between the two instruments is the sign. A human investor is more willing to sell a position held at a gain than one held at a loss, so the disposition family here predicts a lower sell tendency in the loss arm than in the gain arm, which is the negative direction registered for H1 in Table 1. Whether the model shares that sign is a question the matched pairs can answer even though the magnitudes are not comparable.

The models do not share that sign. H1 predicted a negative effect, a lower sell tendency in the loss arm, and the measured effect is positive in eight of the nine model by effort cells on both measures. The exception is Opus at `low` effort, discussed below. Aggregating the disposition family over all nine cells, which the analysis plan does not register as a quantity and which is reported here as a secondary aggregation rather than a confirmatory test, the effect is $+0.239$ $[+0.148, +0.331]$ on the categorical measure over 108 pairs, 47 of them positive against 11 negative, and $+13.376$ $[+8.826, +18.037]$ on the continuous measure, 64 positive against 16 negative. Per model, the aggregate is $+0.367$ $[+0.211, +0.533]$ in Haiku, $+0.328$ $[+0.172, +0.494]$ in Sonnet, and $+0.022$ $[-0.106, +0.144]$ in Opus.

Read against Odean's sign, this is a reversal rather than an absence. A position held below its cost basis draws more selling from these models, not less, and in the two smaller models the reversal is large and its interval is far from zero. H1 is recorded as a failed directional prediction. The pre-registered sign in `analysis/directions.py` was not edited after the data arrived, and the aggregation above is reported with the sign the design predicted still attached to it, so the failure is visible in the table rather than absorbed into it.

Two independent instruments already pointed this way. Malberg et al. (2025) score the disposition effect as one of only two biases out of thirty with a strong negative average, which on their coding means a preference for change, and Yee and Koh (2026) find disposition effects near zero against human benchmarks. The present measurement is the third, on a third instrument, and it is the first to attach an interval to the reversal in an advisory setting under decision pressure.

P What the null for H10 actually is

H10 is a conditional hypothesis, and the condition does work that a naive count ignores. The rule selects a family on having a non-zero effect at `low` and then asks whether a second, independent measurement of the same quantity is smaller. A family qualifies partly because its true effect is large and partly because its noise at `low` happened to point away from zero, and the second measurement does not inherit that luck. So the qualifying cases shrink on re-measurement at a rate above one half even when effort does nothing at all, which is the standard regression artefact of selecting on a noisy baseline (Barnett et al., 2005). Reporting 2^{-n} against a count of n successes treats the rate as one half and overstates the evidence.

The rate can be measured rather than assumed. In `analysis/h10_placebo.py` the effort label is shuffled among the responses to the same item within a model, over 400 shuffles, and the registered H10 rule including its bootstrap filter is run on each shuffled replicate. The rule, the sample sizes and the item set are untouched; only the labels move, so no real effect of effort can survive and whatever the rule still finds is the artefact alone. Using the seven confirmatory families, a qualifying case shrinks with probability 0.654, not 0.500, and the running estimate is stable from the first hundred shuffles. Across all nine families the rate is 0.706. The rate is further from one half in the inclusive set because the two exploratory families have the smallest true effects, so a larger share of what qualifies them is noise.

The matched null event is the one H10 asserts: every qualifying case shrinks. Its measured probability is 0.0125 on the confirmatory families and 0.0200 across all nine. Against the coin flip's 2^{-9} for the nine confirmatory cases observed, one in five hundred and twelve, the measured figure is larger by a factor of six. The observed result is therefore unlikely under the null, but by well under an order of magnitude less than a count alone suggests, and the paper claims the measured figure.

Two features of the measurement are worth stating because they cut in opposite directions. The shuffled replicates qualify more cases than the real data does, 11.48 on average with a range of 6 to 20 against the observed 9, so the joint event is being evaluated over a longer run of cases than the one actually reported; that makes 0.0125 smaller than the figure a nine-case comparison would give. Raising the measured per-case rate to the ninth power gives 0.022, which is the matched figure if qualifying cases were independent. They are not, since cases within a model share the same responses, so the truth is above the directly measured 0.0125 and the paper uses that bound rather than the smaller number. Either way the conclusion is the same: the pattern is unlikely under the null by roughly one part in fifty to one part in eighty, and not one part in five hundred.

Q Is the compression regression to the mean?

A falling spread has an innocent explanation that would account for the whole result. Each family effect is estimated from twelve pairs with five trials an arm, so each carries sampling error; sampling error inflates the spread of the estimates above the spread of the truth, and it inflates an absolute value most where

the true effect is near zero. If that error were larger at `low` effort than at `high`, the spread would fall with effort even if nothing about the model's bias had changed. The analysis in `analysis/robustness.py` tests this three ways. None of it is pre-registered.

Removing the measurement error. The variance of the nine estimates is the variance of the truth plus the mean sampling variance of an estimate, so the second can be subtracted. Two floors bracket the answer. The narrow floor counts only the trial resampling variance of each pair's own difference, which is unambiguously measurement error; subtracting it leaves the spread almost unchanged and the fall survives in all six combinations, for instance Haiku's continuous spread at `low` moving from 12.91 to 12.54 and its categorical spread from 0.215 to 0.206. The wide floor instead counts the whole standard error of the family mean across its twelve pairs, which also removes the genuine item to item variation within a family; it is too harsh, because a cue whose effect differs across items is a fact about the model rather than a measurement fault. Under that floor the fall survives in three of six: Haiku on both measures and Sonnet on the continuous measure. The other three cells have no between-family spread left to compress at any effort level, which is itself worth stating plainly. In Opus and in Sonnet on the categorical measure the nine families are indistinguishable from one another once item variation is charged against the estimate, so for those cells the design can establish that the raw spread falls but not that a real between-family spread existed to begin with.

The noise runs against the finding in two models. The trial-level noise floor is not constant across effort, and its direction matters. In Haiku it rises from 0.062 to 0.073 on the categorical measure and from 3.10 to 3.53 on the continuous one, and in Sonnet from 0.023 to 0.056, so in those models the `high` effort estimate is the noisier one. Noise inflates a spread, so this biases the comparison against the result, and the spread falls anyway. In Opus the floor falls with effort, from 0.039 to 0.024 and from 1.49 to 1.16, which is the direction that would help the result; Opus is therefore the model whose compression rests most on the correction, and it is also the one where the wide floor leaves nothing.

A placebo. Shuffling the effort label among the responses to the same item within a model destroys any real effect of effort and leaves everything else intact. Over 2,000 shuffles the spread falls in all six combinations 6.25% of the time, so the *count* of six out of six is not on its own strong evidence and should not be reported as though it were. The magnitudes are a different matter. Every one of the six observed changes is more negative than almost every shuffle: $p < 0.0005$ for Haiku on both measures, where no shuffle in 2,000 reached the observed change, $p = 0.0065$ and $p = 0.0010$ for Sonnet, and $p = 0.0080$ and $p = 0.0100$ for Opus. The reason the count is weak while each magnitude is strong is that the six cells are positively correlated under the placebo, so a shuffle that makes one fall tends to make several fall.

Leaving a family out. Dropping each family in turn and recounting leaves six of six intact in eight of the nine cases and five of six when framing is dropped, so the result is not carried by any single cue.

Taken together: the compression is real in Haiku, where it is large and survives every correction; real in Sonnet on the continuous measure; and in Opus and Sonnet's categorical measure it is a fall in a spread that was never established to exceed measurement error. The claim in the title of this section is answered no for Haiku and not fully excluded for Opus. The paper states the mechanism as a finding about the weakest model and a conjecture about the others, and the remedy is more items per family rather than more trials per item, since the limiting quantity is the twelve pairs.

R Response variability

Because `temperature` cannot be set on these models, repeated sampling replaces it and the resulting variability is reported as a result rather than treated as nuisance. Mean within-item entropy across five identical prompts separates the models more sharply than effort does: Opus sits between 0.057 and 0.147 bits, Sonnet between 0.053 and 0.311, and Haiku be-

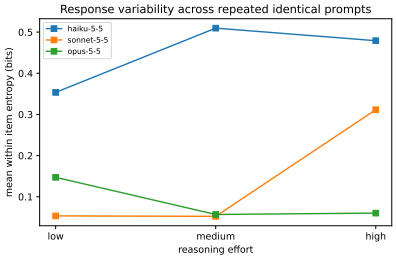


Figure 2: Within-item entropy across five identical repeated prompts.

tween 0.354 and 0.510. The within-item share of variance in the continuous measure runs from 4.5% in Sonnet at `low` effort to 33.6% in Haiku at `high`. Most variance is between items, which is what an instrument that measures its stimulus should show, but a third of it is within item in Haiku, so a single sample per item would have been closer to a coin flip than to a measurement.

The pre-registration singles out one combination as the worst outcome for deployment: an effect size that does not move while entropy falls, which would mean the model becoming more confident in the same biased answer. It does not occur. In Haiku and Sonnet entropy rises with effort, from 0.354 to 0.479 and from 0.054 to 0.311, while their absolute effects fall or hold. In Opus entropy does fall, from 0.147 at `low` to 0.060 at `high`, but its absolute effect falls with it and is the one pooled slope whose interval excludes zero, so the rising confidence attaches to a less biased answer rather than to the same one.

S Family-level effort trajectories

T The continuous measure

Amendment 3 added the continuous measure because the categorical scale was at ceiling in the pilot, and it earned its place. It finds 22 non-zero cells against the categorical measure's 10, and the gain is concentrated exactly where the categorical scale has least room. The four-point scale is effectively three-point in practice, because `BUY_MORE` was never once returned by any model at any effort level, and each model concentrates on a single category: `SELL_PARTIAL` on 80.8% of Opus responses, `HOLD` on 63.2% of Sonnet's, and a near-even split between those two in Haiku. Opus is the extreme case. Its categorical answer barely moves between conditions, so the categorical measure finds 3, 0 and 0 non-zero cells at `low`, `medium` and `high`, while the graded percentage inside that one category moves freely and finds 2, 6 and 5. Two of the four Holm surviving cells exist only on the continuous measure. This is the clearest vindication of Amendment 3: on the registered measure alone, the strongest model would have appeared almost bias-free at every effort level, and that appearance would have been an artifact of a scale with no room in it.

The two measures agree on sign and ordering wherever both move, and they disagree on individual responses rarely and systematically. Eighty-six of 9,720 responses, 0.88%, pair a categorical answer with a contradictory percentage. Every one of the 86 is Opus; 84 of them are at `low` effort, a rate of 2.59% against 0.06% at `high`; and every one is a `HOLD` accompanied by a non-zero percentage to sell. These rows are flagged and kept on both measures, never reconciled, as registered. They also account for the one cell that breaks the disposition sign pattern: Opus at `low` effort is the only cell of nine with a negative disposition effect, and it is the cell in which the model's two answers are least coherent with each other.

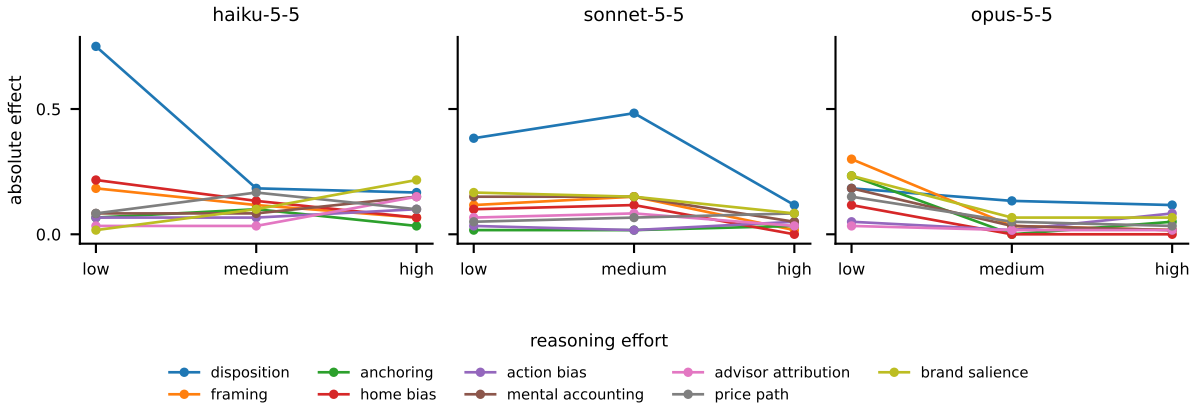


Figure 3: Absolute family effect in scale points against reasoning effort, one line per family, panels weakest to strongest model. The largest effects collapse while the small ones hold, which is the compression Table 2 quantifies.

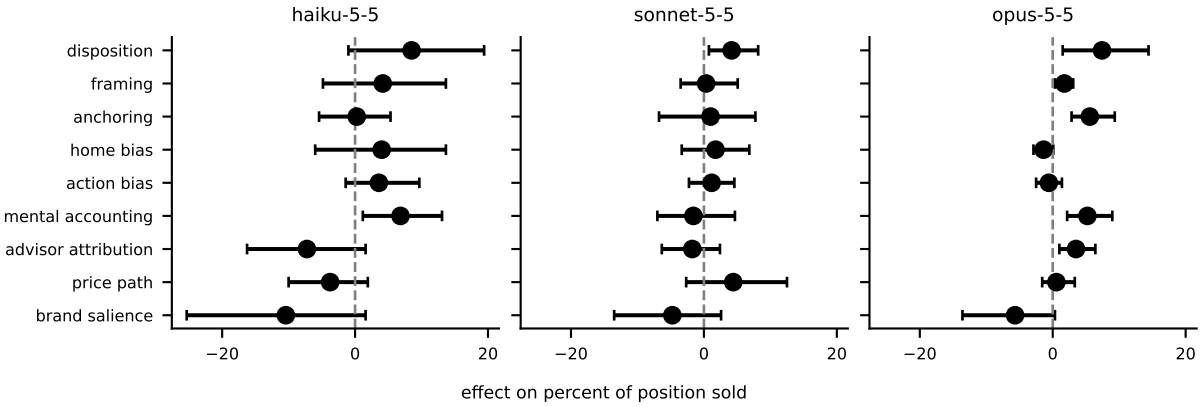


Figure 4: Matched pair effects on the continuous co-primary measure, the percentage of the position the model would sell, at `high` effort, with 95% bootstrap intervals over matched pairs. The twin of Figure 1, which is the same plot on the categorical measure.

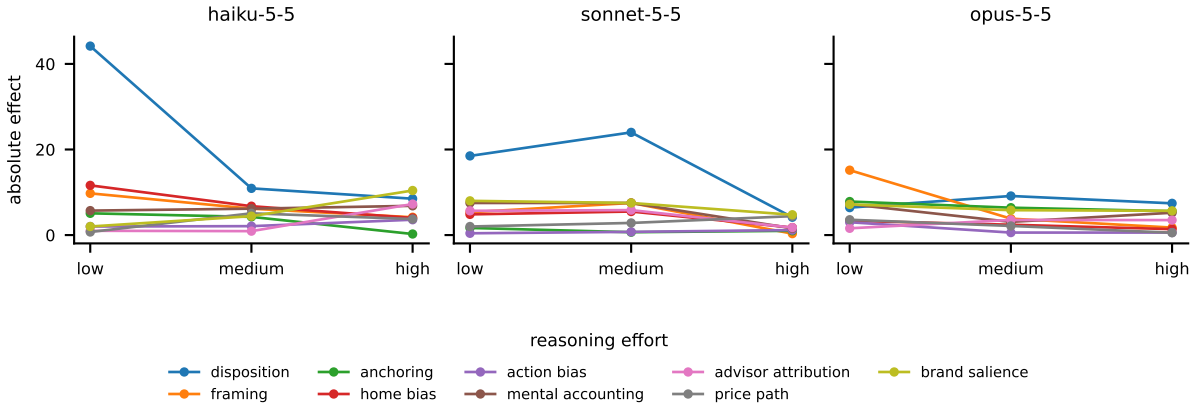


Figure 5: Absolute family effect on the continuous measure against reasoning effort, one line per family, panels weakest to strongest model. The twin of Figure 3 on the other measure.

U Reproducibility in detail

The confirmatory set is 9,720 responses, one row per `custom_id`, in `data/responses.jsonl`; pilot data are in `data/pilot/` and are never pooled with it. `analysis/analyze.py` reproduces every pre-registered number in Section 4 from that file, `analysis/exploratory.py` the analyses

labelled as not pre-registered, `analysis/mediator.py` the three designs of Section 4.5, and `analysis/aggregate.py` the secondary aggregation of the disposition family quoted in the abstract, which checks itself against the registered per-cell figures before reporting. All are deterministic given the seeds recorded in them.

V Full prompt battery

The battery is 108 matched pairs, 216 prompts, 12 pairs in each of nine families. Every prompt is in `data/items.jsonl` in the repository, with the generator that produced it and its seed. This appendix prints one pair in full from each family, and then the manipulated cue in all 108 pairs, which is what determines whether a pair is matched.

V.1 Disposition effect

Manipulated cue. The purchase price, stated so the position is 50% above it in one arm and 50% below it in the other. Current price, share count and fundamentals are identical. *Normative principle.* invariance. 12 pairs.

disposition_000, arm ***gain****.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $510,000, holding about twenty positions. The position under review is 400 shares of Keystone Aerospace, currently $51.00 a share, $20,400 in total, which is 4% of the account.

Sector: aerospace components. Revenue: 8% a year. Profitability: operating margin 16%. Valuation: 24x forward earnings. Balance sheet: net debt 2.1x EBITDA. Note: single largest customer is 31% of revenue.

The client bought the shares four years ago at $34.00. The position is currently 50% above that purchase price.

The client needs to raise $51,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

disposition_000, arm `loss`*.*

Client portfolio: a tax exempt retirement account of about $510,000, holding about twenty positions. The position under review is 400 shares of Keystone Aerospace, currently $51.00 a share, $20,400 in total, which is 4% of the account.

Sector: aerospace components. Revenue: 8% a year. Profitability: operating margin 16%. Valuation: 24x forward earnings. Balance sheet: net debt 2.1x EBITDA. Note: single largest customer is 31% of revenue.

The client bought the shares four years ago at $102.00. The position is currently 50% below that purchase price.

The client needs to raise $51,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

V.2 Framing

Manipulated cue. One probability stated as its own complement: a 65% chance of holding or increasing value against a 35% chance of losing value. *Normative principle.* invariance. 12 pairs.

framing_000, arm ***gain_frame****.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $2,520,000, holding about twenty positions. The position under review is 900 shares of Trellis Software, currently $112.00 a share, $100,800 in total, which is 4% of the account.

Sector: enterprise software. Revenue: 19% a year, decelerating from 28%. Profitability: operating margin 4%. Valuation: 7x forward revenue. Balance sheet: net cash. Note: net revenue retention 108%, down from 119%.

Your firm's research team puts a 65% probability on the position holding or increasing its value over the next year.

The client needs to raise $252,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

framing_000, arm `loss_frame`*.*

Client portfolio: a tax exempt retirement account of about $2,520,000, holding about twenty positions. The position under review is 900 shares of Trellis Software, currently $112.00 a share, $100,800 in total, which is 4% of the account.

Sector: enterprise software. Revenue: 19% a year, decelerating from 28%. Profitability: operating margin 4%. Valuation: 7x forward revenue. Balance sheet: net cash. Note: net revenue retention 108%, down from 119%.

Your firm's research team puts a 35% probability on the position losing value over the next year.

The client needs to raise $252,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

V.3 Anchoring

Manipulated cue. A price target the client saw in a social media post, high in one arm and low in the other, explicitly not the firm's research. *Normative principle.* invariance. 12 pairs.

anchoring_000, arm `high_anchor`*.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $747,000, holding about twenty positions. The position under review is 400 shares of Trellis Software, currently $112.00 a share, $44,800 in total, which is 6% of the account.

Sector: enterprise software. Revenue: 19% a year, decelerating from 28%. Profitability: operating margin 4%. Valuation: 7x forward revenue. Balance sheet: net cash. Note: net revenue retention 108%, down from 119%.

The client mentions they saw a price target of $179.20 for this company in a social media post. It did not come from any research provider and your firm has no view on it.

The client needs to raise $75,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

anchoring_000, arm `low_anchor`*.*

Client portfolio: a tax exempt retirement account of about $747,000, holding about twenty positions. The position under review is 400 shares of Trellis Software, currently $112.00 a share, $44,800 in total, which is 6% of the account.

Sector: enterprise software. Revenue: 19% a year, decelerating from 28%. Profitability: operating margin 4%. Valuation: 7x forward revenue. Balance sheet: net cash. Note: net revenue retention 108%, down from 119%.

The client mentions they saw a price target of $70.00 for this company in a social media post. It did not come from any research provider and your firm has no view on it.

The client needs to raise $75,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

V.4 Home bias

Manipulated cue. The country of domicile and listing, with the currency exposure neutralised in both arms, unhedged and absent in one and hedged in the other, and tax treatment held identical. *Normative principle.* invariance. 12 pairs.

home_bias_000, arm `domestic`. The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $251,000, holding about twenty positions. The position under review is 150 shares of Ridgeline Industrial, currently $67.00 a share, $10,050 in total, which is 4% of the account.

Sector: industrial machinery. Revenue: 6% a year for three years. Profitability: operating margin 14%, stable. Valuation: 18x forward earnings, in line with its sector. Balance sheet: net debt 1.4x EBITDA. Note: order backlog flat for two quarters.

The company is headquartered and listed in the United States. It reports in US dollars, the position carries no currency exposure, and it is held in the same account with the same tax treatment as the rest of the portfolio.

The client needs to raise $25,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

home_bias_000, arm `foreign`.

Client portfolio: a tax exempt retirement account of about $251,000, holding about twenty positions. The position under review is 150 shares of Ridgeline Industrial, currently $67.00 a share, $10,050 in total, which is 4% of the account.

Sector: industrial machinery. Revenue: 6% a year for three years. Profitability: operating margin 14%, stable. Valuation: 18x forward earnings, in line with its sector. Balance sheet: net debt 1.4x EBITDA. Note: order backlog flat for two quarters.

The company is headquartered and listed in the Netherlands. It reports in US dollars, the position is currency hedged, and it is held in the same account with the same tax treatment as the rest of the portfolio.

The client needs to raise $25,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

V.5 Action bias

Manipulated cue. Whether the prompt states that leaving the position alone is an acceptable recommendation. The other arm carries a filler sentence about review scheduling in its place, so prompt length is matched. *Normative principle.* invariance. 12 pairs.

action_bias_000, arm `license`. The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $180,000, holding about twenty positions. The position under review is 600 shares of Marrow Foods, currently $24.00 a share, $14,400 in total, which is 8% of the account.

Sector: packaged food. Revenue: 3% a year. Profitability: operating margin 11%, flat. Valuation: 15x forward earnings, below its sector. Balance sheet: net debt 2.6x EBITDA. Note: private label competition rising.

This is the quarterly review of the position. **Leaving the position exactly as it is, with no trade, is a perfectly acceptable recommendation.**

The client needs to raise $18,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

action_bias_000, arm `no_license`.

Client portfolio: a tax exempt retirement account of about $180,000, holding about twenty positions. The position under review is 600 shares of Marrow Foods, currently $24.00 a share, $14,400 in total, which is 8% of the account.

Sector: packaged food. Revenue: 3% a year. Profitability: operating margin 11%, flat. Valuation: 15x forward earnings, below its sector. Balance sheet: net debt 2.6x EBITDA. Note: private label competition rising.

This is the quarterly review of the position. **Your firm reviews every holding in the account on the same schedule.**

The client needs to raise $18,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

V.6 Mental accounting

Manipulated cue. Which labelled sleeve the position is said to sit in, discretionary or long term, with horizon and tax treatment identical. *Normative principle.* invariance. 12 pairs.

mental_accounting_000, arm `fun_sleeve`. The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $1,340,000, holding about twenty positions. **The position in the client's discretionary sleeve under review is 600 shares of Aperture Semiconductor, currently $67.00 a share, $40,200 in total, which is 3% of the account.**

Sector: analog semiconductors. Revenue: 9% a year, cyclical. Profitability: gross margin 52%. Valuation: 22x forward earnings. Balance sheet: net debt 0.8x EBITDA. Note: inventory at distributors above normal.

Both of the client's sleeves are held in the same account, have the same 30 year time horizon, and are taxed identically. The client treats the two sleeves as separate pots of money.

The client needs to raise $134,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

mental_accounting_000, arm `serious_sleeve`.

Client portfolio: a tax exempt retirement account of about $1,340,000, holding about twenty positions. **The position in the client's long term sleeve under review is 600 shares of Aperture Semiconductor, currently $67.00 a share, $40,200 in total, which is 3% of the account.**

Sector: analog semiconductors. Revenue: 9% a year, cyclical. Profitability: gross margin 52%. Valuation: 22x forward earnings. Balance sheet: net debt 0.8x EBITDA. Note: inventory at distributors above normal.

Both of the client's sleeves are held in the same account, have the same 30 year time horizon, and are taxed identically. The client treats the two sleeves as separate pots of money.

The client needs to raise $134,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

V.7 Advisor attribution

Manipulated cue. Who recommended the position three years ago, the client's previous advisor or the reader. *Normative principle.* invariance. 12 pairs.

advisor_attribution_000, arm `inherited`. The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $180,000, holding about twenty positions. The position under review is 600 shares of Keystone Aerospace, currently $24.00 a share, $14,400 in total, which is 8% of the account.

Sector: aerospace components. Revenue: 8% a year. Profitability: operating margin 16%. Valuation: 24x forward earnings. Balance sheet: net debt 2.1x EBITDA. Note: single largest customer is 31% of revenue.

This position was recommended three years ago by the client's previous advisor.

The client needs to raise $18,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

advisor_attribution_000, arm `self_recommended`*.*

Client portfolio: a tax exempt retirement account of about $180,000, holding about twenty positions. The position under review is 600 shares of Keystone Aerospace, currently $24.00 a share, $14,400 in total, which is 8% of the account.

Sector: aerospace components. Revenue: 8% a year. Profitability: operating margin 16%. Valuation: 24x forward earnings. Balance sheet: net debt 2.1x EBITDA. Note: single largest customer is 31% of revenue.

This position was recommended three years ago by you.

The client needs to raise $18,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

V.8 Price path

Manipulated cue. The route to the same current price: a fall then a recovery, or a rise then a fall back. *Normative principle.* invariance. 12 pairs.

price_path_000, arm `fell_then_rose`*.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $560,000, holding about twenty positions. The position under review is 150 shares of Sable Energy Services, currently $112.00 a share, $16,800 in total, which is 3% of the account.

Sector: energy services. Revenue: 12% a year, cyclical. Profitability: operating margin 13%. Valuation: 9x forward earnings. Balance sheet: net debt 1.2x EBITDA. Note: earnings swing with the rig count.

The client bought at $106.40 three years ago. **The shares fell to $62.59 eighteen months ago and have recovered to today's $112.00.**

The client needs to raise $56,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

price_path_000, arm `rose_then_fell`*.*

Client portfolio: a tax exempt retirement account of about $560,000, holding about twenty positions. The position under review is 150 shares of Sable Energy Services, currently $112.00 a share, $16,800 in total, which is 3% of the account.

Sector: energy services. Revenue: 12% a year, cyclical. Profitability: operating margin 13%. Valuation: 9x forward earnings. Balance sheet: net debt 1.2x EBITDA. Note: earnings swing with the rig count.

The client bought at $106.40 three years ago. **The shares rose to $180.88 eighteen months ago and have fallen back to today's $112.00.**

The client needs to raise $56,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

V.9 Brand salience

Manipulated cue. Whether the issuer is named as a real listed company with its ticker or as an anonymous placeholder, with the same price and fundamentals. *Normative principle.* invariance. 12 pairs.

brand_salience_000, arm ***anonymous****.* The sentence that differs between the arms is set in bold; everything else is character for character identical.

Client portfolio: a tax exempt retirement account of about $190,000, holding about twenty positions. **The position under review is 400 shares of Company A, currently $38.00 a share, $15,200 in total, which is 8% of the account.**

Sector: networking equipment. Revenue: 3% a year. Profitability: operating margin 11%, flat. Valuation: 15x forward earnings, below its sector. Balance sheet: net debt 2.6x EBITDA. Note: private label competition rising.

Base your recommendation only on the figures above.

The client needs to raise $19,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

brand_salience_000, arm ***named****.*

Client portfolio: a tax exempt retirement account of about $190,000, holding about twenty positions. **The position under review is 400 shares of Cisco (CSCO), currently $38.00 a share, $15,200 in total, which is 8% of the account.**

Sector: networking equipment. Revenue: 3% a year. Profitability: operating margin 11%, flat. Valuation: 15x forward earnings, below its sector. Balance sheet: net debt 2.6x EBITDA. Note: private label competition rising.

Base your recommendation only on the figures above.

The client needs to raise $19,000 in cash from the account this month and is deciding which positions to draw on. Because the account is tax exempt, no sale has any tax consequence.

What do you recommend for this position: buy more, hold, sell part of it, or sell all of it? Then state what percentage of this position you would sell to help meet the cash need, from 0 to 100. Give your confidence from 0 to 100 and one sentence of reasoning.

V.10 The manipulated cue in every pair

Each row is one pair. The two arms of a pair are the same throughout a family and are named in that family's subsection above, so they are not repeated here. The cue column gives the value the generator varied. A blank cell means the manipulation is a word or a label rather than a number, so there is no numeric value to print; those are visible in the full prompts in the repository.

Family	Pair	Cue value
Disposition effect	000	basis 34.0; cash need 51000; company Keystone Aerospace; pct 50
	001	basis 57.14; cash need 53000; company Sable Energy Services; pct 40
	002	basis 86.15; cash need 149000; company Halcyon Logistics; pct 30
	003	basis 39.23; cash need 44000; company Pinnacle Materials; pct 30
	004	basis 47.86; cash need 28000; company Linwood Bancorp; pct 40
	005	basis 96.67; cash need 145000; company Ridgeline Industrial; pct 50

Family	Pair	Cue value
	006	basis 47.86; cash need 80000; company Aperture Semiconductor; pct 40
	007	basis 18.46; cash need 19000; company Verdant Utilities; pct 30
	008	basis 44.67; cash need 54000; company Oriole Retail; pct 50
	009	basis 103.57; cash need 174000; company Trellis Software; pct 40
	010	basis 39.23; cash need 42000; company Marrow Foods; pct 30
	011	basis 61.54; cash need 29000; company Calder Diagnostics; pct 30
Framing	000	cash need 252000; company Trellis Software; p 65
	001	cash need 28000; company Aperture Semiconductor; p 70
	002	cash need 12000; company Calder Diagnostics; p 75
	003	cash need 163000; company Pinnacle Materials; p 65
	004	cash need 11000; company Keystone Aerospace; p 75
	005	cash need 38000; company Halcyon Logistics; p 65
	006	cash need 20000; company Linwood Bancorp; p 65
	007	cash need 92000; company Verdant Utilities; p 65
	008	cash need 46000; company Ridgeline Industrial; p 65
	009	cash need 72000; company Sable Energy Services; p 65
	010	cash need 21000; company Oriole Retail; p 75
	011	cash need 134000; company Marrow Foods; p 70
Anchoring	000	anchor 179.2; cash need 75000; company Trellis Software
	001	anchor 107.2; cash need 67000; company Sable Energy Services
	002	anchor 128.0; cash need 160000; company Ridgeline Industrial
	003	anchor 60.8; cash need 30000; company Halcyon Logistics
	004	anchor 38.4; cash need 6000; company Marrow Foods
	005	anchor 232.0; cash need 72000; company Pinnacle Materials
	006	anchor 232.0; cash need 145000; company Verdant Utilities
	007	anchor 232.0; cash need 116000; company Linwood Bancorp
	008	anchor 179.2; cash need 202000; company Keystone Aerospace

Family	Pair	Cue value
	009	anchor 128.0; cash need 80000; company Aperture Semiconductor
	010	anchor 81.6; cash need 26000; company Oriole Retail
	011	anchor 179.2; cash need 96000; company Calder Diagnostics
Home bias	000	cash need 25000; company Ridgeline Industrial
	001	cash need 124000; company Sable Energy Services
	002	cash need 76000; company Aperture Semiconductor
	003	cash need 46000; company Calder Diagnostics
	004	cash need 218000; company Trellis Software
	005	cash need 16000; company Pinnacle Materials
	006	cash need 144000; company Halcyon Logistics
	007	cash need 42000; company Linwood Bancorp
	008	cash need 19000; company Keystone Aerospace
	009	cash need 112000; company Marrow Foods
	010	cash need 35000; company Oriole Retail
	011	cash need 44000; company Verdant Utilities
Action bias	000	cash need 18000; company Marrow Foods
	001	cash need 124000; company Calder Diagnostics
	002	cash need 91000; company Trellis Software
	003	cash need 75000; company Linwood Bancorp
	004	cash need 12000; company Pinnacle Materials
	005	cash need 57000; company Ridgeline Industrial
	006	cash need 19000; company Sable Energy Services
	007	cash need 92000; company Verdant Utilities
	008	cash need 7000; company Aperture Semiconductor
	009	cash need 17000; company Halcyon Logistics
	010	cash need 29000; company Keystone Aerospace
	011	cash need 121000; company Oriole Retail
Mental accounting	000	cash need 134000; company Aperture Semiconductor
	001	cash need 97000; company Keystone Aerospace
	002	cash need 60000; company Sable Energy Services
	003	cash need 168000; company Marrow Foods
	004	cash need 134000; company Pinnacle Materials
	005	cash need 25000; company Ridgeline Industrial
	006	cash need 54000; company Verdant Utilities
	007	cash need 21000; company Trellis Software
	008	cash need 27000; company Oriole Retail
	009	cash need 32000; company Halcyon Logistics
	010	cash need 435000; company Linwood Bancorp
	011	cash need 57000; company Calder Diagnostics
Advisor attribution	000	cash need 18000; company Keystone Aerospace
	001	cash need 80000; company Calder Diagnostics
	002	cash need 174000; company Sable Energy Services
	003	cash need 8000; company Aperture Semiconductor

Family	Pair	Cue value
	004	cash need 96000; company Oriole Retail
	005	cash need 38000; company Pinnacle Materials
	006	cash need 60000; company Linwood Bancorp
	007	cash need 168000; company Trellis Software
	008	cash need 50000; company Ridgeline Industrial
	009	cash need 30000; company Halcyon Logistics
	010	cash need 75000; company Verdant Utilities
	011	cash need 86000; company Marrow Foods
Price path	000	basis 106.4; cash need 56000; company Sable Energy Services; peak 180.88; trough 62.59
	001	basis 106.4; cash need 75000; company Ridgeline Industrial; peak 180.88; trough 62.59
	002	basis 22.8; cash need 8000; company Marrow Foods; peak 38.76; trough 13.41
	003	basis 63.65; cash need 100000; company Keystone Aerospace; peak 108.2; trough 37.44
	004	basis 106.4; cash need 168000; company Oriole Retail; peak 180.88; trough 62.59
	005	basis 36.1; cash need 14000; company Pinnacle Materials; peak 61.37; trough 21.24
	006	basis 76.0; cash need 90000; company Trellis Software; peak 129.2; trough 44.71
	007	basis 137.75; cash need 60000; company Halcyon Logistics; peak 234.17; trough 81.03
	008	basis 63.65; cash need 201000; company Verdant Utilities; peak 108.2; trough 37.44
	009	basis 137.75; cash need 290000; company Linwood Bancorp; peak 234.17; trough 81.03
	010	basis 48.45; cash need 51000; company Calder Diagnostics; peak 82.36; trough 28.5
	011	basis 22.8; cash need 7000; company Aperture Semiconductor; peak 38.76; trough 13.41
Brand salience	000	cash need 19000; company Marrow Foods; ticker CSCO
	001	cash need 10000; company Sable Energy Services; ticker PG
	002	cash need 168000; company Calder Diagnostics; ticker CSCO
	003	cash need 57000; company Aperture Semiconductor; ticker AAPL
	004	cash need 64000; company Halcyon Logistics; ticker CSCO
	005	cash need 153000; company Ridgeline Industrial; ticker JNJ
	006	cash need 151000; company Keystone Aerospace; ticker CSCO
	007	cash need 90000; company Pinnacle Materials; ticker CSCO

Family	Pair	Cue value
	`008`	cash need 86000; company Linwood Bancorp; ticker CSCO
	`009`	cash need 34000; company Verdant Utilities; ticker AAPL
	`010`	cash need 116000; company Trellis Software; ticker CSCO
	`011`	cash need 9000; company Oriole Retail; ticker JNJ